System Biology - Project 1 - KCWM - Team 8 - Task C

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1 Introduction

In the classical Fisher's Geometric Model, mutations are assumed to be isotropic - occurring with equal probability in ay direction in phenotypic space. While this serves as a robust train model, biological evidence suggests that mutation bias can evolve as an adaptive trait. In environments characterized by sustained directional selection, populations that can "align" their mutational input with the direction of environment shift may fit better to the changing environment.

In our study we investigated whether we can observe how environment enforces more focus on directional changes in specific scenario. We analyze Adaptive Directional Mutation Strategy: Every *individual* independently tunes its directional-to-isotropic mutation ratio, separately for each dimension. Our goal is to find out if such "metaparameters" of an *individual* can influence its fitness, and if they can fit into the environmental condition.

We test mutation strategy in environment characterized by steady, fast drift. The drift could happen in all dimensions at a similar speed, or in only one direction.

We expect that, scenario with fast, steady, drift across one selected dimension will enforce population to adapt their directional ratio to focus on environment-driven changes, whereas on other dimensions ratios should have high variance, since they do not matter this much.

2 Methods

2.1 Model

We utilize a simulation of Fisher's Geometric Model where a population of size N exist in a n-dimensional phenotypic space. The fitness of an individual is determined by a Gaussian function of phenotypic distance from optimum.

$$\varphi(\mathbf{p}, \boldsymbol{\alpha}) = \exp\left(-\frac{\|\mathbf{p} - \boldsymbol{\alpha}\|^2}{2\sigma^2}\right)$$

In our experiment we consider population that reproduces asexually.

2.2 Mutation Strategies

During each iteration individuals mutate, following one of the **mutation strategies**.

Parameters μ, μ_c, ξ are set globally for the entire population and remain constant throughout entire simulation.

2.2.1 Adaptive Directional Mutation

For this scenario, every *individual*, except its phenotype, carries also vector $\lambda \in R^n$. λ does not influence individual's fitness directly. They were initialized from uniform distribution $Unif(0, 1)$.

This mutation strategy was implemented in our code in `strategies/mutation.py` as `class WeightedShiftsMutation(UnifiedMutations)` class. Every mutation consist of *directional* and *isotropic* components.

$$directional_{i+1} = \alpha_{i+1} - \alpha_i$$

isotropic component is calculated in the same way as in **Isotropic Mutation**. It follows *two-level Bernoulli process*. Each individual has probability μ to mutate at all, with probability $1 - \mu$ it passes unchanged. For each mutant, every phenotypic feature $i \in 1, ..., n$ mutates independently with probability μ_c. The mutation is drawn from a zero-mean isotropic Gaussian:

$$\Delta p_i \sim \mathcal{N}(0,\, \xi^2), \qquad \text{independently for each } i.$$

Phenotype change Δp_i is calculated using formula:

$$\Delta p_i = \lambda \circ \mathit{directional} + (\mathbf{1}_n - \lambda) \circ \mathit{isotropic}$$

where $\circ$ denotes element-wise multiplication.

After individual's phenotype mutation, vector λ also mutates, in a way that keeps its values in range $[0, 1]$.

$$\lambda_{i+1} = \mathit{min}(1, \mathit{max}(0,\ \lambda_i + N(0, 0.1)))$$

2.3 Environment

After reproduction, the optimal phenotype is updated according to linear drift with additive Gaussian noise:

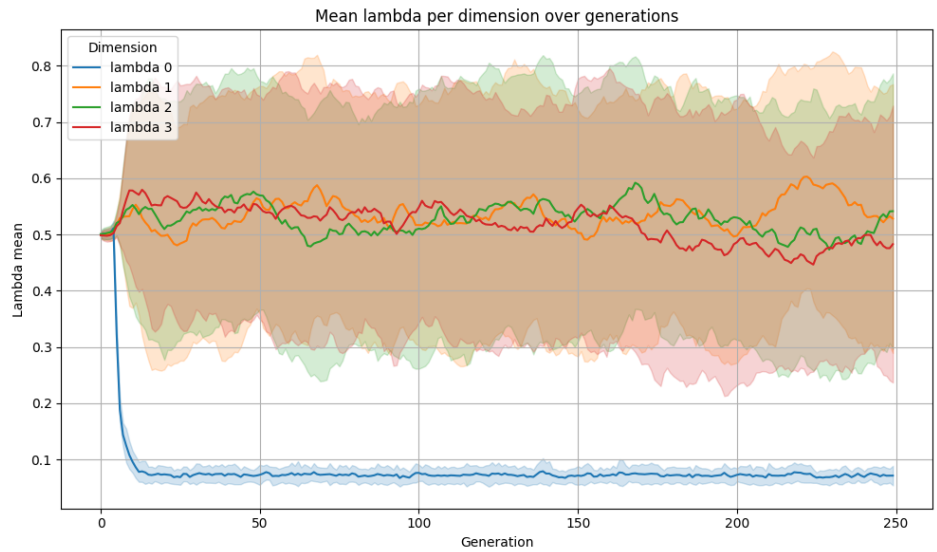
$$\boldsymbol{\alpha}(t+1) = \boldsymbol{\alpha}(t) + \boldsymbol{\varepsilon}_t, \qquad \boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{c},\, \delta^2 \mathbf{I}_n),$$

where

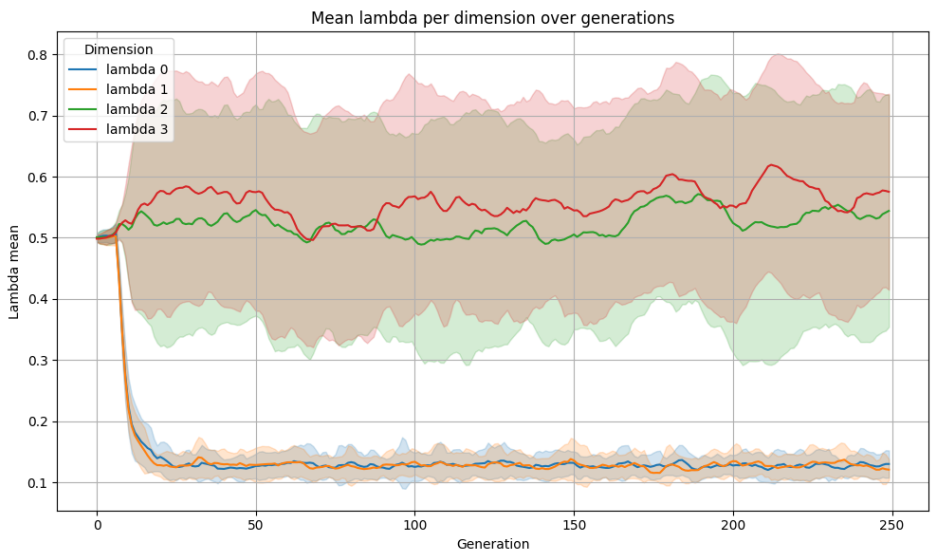
- $\mathbf{c} \in R^n$ (‘c‘) is the **mean drift vector** — the deterministic component of environmental change per generation,
- $\delta \geq 0$ (‘delta‘) is the **stochastic volatility** — the standard deviation of isotropic fluctuations around $\mathbf{c}$.

In our experiments we utilized environment where $n = 4$. To control dimensions of drift we utilized parameter c (lower value at position i means slower drift in i-th dimension. We run experiments with 40 replicates, $N = 5000$ and max 250 generations. The other parameters were chosen in a way that avoided extinction of populations. The final experiment results are available in `relevant_experiment_results/`.

3 Results



(a) Mean and std of lambda in each dimension. $c = [0.25, 1e-7, 1e-7, 1e-7]$



(b) Mean and std of lambda in each dimension. $c = [0.125, 0.125, 1e-7, 1e-7]$

We can observe that stronger drift in dimension i forces almost constant value of λ_i and reduces variance.

4 Discussion

We find it surprising that λ diverged to ~ 0.1, we were expecting it to be closer to 1, since that would mean exact copy of the last environmental shift. Values of λ on positions that don't influence the phenotype this much was expected, it shows that λ on those positions do not matter to individual's fitness. In our opinion the most relevant observation is fact that we can clearly observe that λ can "evolve" with the population even if they are not directly part of the phenotype.